Qianyu Dong

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Education

University of California, Santa Cruz — **September 2022–Present**
PhD candidate in Statistical Science — Santa Cruz, CA

Wake Forest University — **August 2020–May 2022**
M.S. in Mathematical Statistics — Winston-Salem, NC

University of Nebraska–Lincoln — **August 2016–August 2019**
B.S. in Mathematics — Lincoln, NE
B.S. in Business Administration, Supply Chain Management, High Distinction

Industry Experience

Ipsen Biopharmaceuticals — June 2026 – August 2026
Oncology Biostatistics Intern — *Cambridge, MA*
Supervisors: Anna Hu, Jianming Wang

- Developed Bayesian adaptive dose-finding methods for oncology combination trials by extending POCRM with latent toxicity orderings and drug-level DLT-informed Mallows priors.
- Built an R Shiny application integrating dose-transition pathways, escalation and safety rules, and automated POCRM reports to support early-phase trial design and protocol development.
- Packaged methods into a reusable R package with automated CI/CD and designed Python integration via a plumber API and Streamlit UI.

Publications

In progress

1. **Dong, Q.**; and Li, Z. R. Design-based cross-validation for comparing small area estimators. (Under review)
2. Kawano, S.; Vedensky, D.; **Dong, Q.**; Pawl, E.; Wang, Q.; Parker, P. A.; Li, Z. R.; and Holan, S. H. Nonprobability samples for small area estimation: A review and comparative simulation study. https://arxiv.org/abs/2608.13673. (Submitted)

Published and accepted

1. Wu, Y.; **Dong, Q.**; Knutson, V.; Xu, J.; Li, Z. R.; and Wakefield, J. (2026). Small area estimation in low- and middle-income countries with the sae4health R Shiny application. *Scientific Reports*. https://doi.org/10.1038/s41598-026-72265-9.
2. Qi, X.; **Dong, Q.**; Pei, Y.; Li, G.; Tang, W.; Hall, B. J.; and Wu, B. (2026). A machine learning approach to identify the gender-specific risk and protective factors of loneliness among adults in Shanghai, China. *Discover Public Health*, *23*(1), 847. https://doi.org/10.1186/s12982-026-02034-6.
3. **Dong, Q.**; Wu, Y.; Li, Z. R.; and Wakefield, J. (2026). Toward a principled workflow for prevalence mapping using household survey data. *Journal of Survey Statistics and Methodology*, *14*(1), 209–237. https://doi.org/10.1093/jssam/smaf048.

4. **Dong, Q.**; Wu, W.; Jiang, Y.; Sui, J.; Tan, C.; and Qi, X. (2025). Machine learning approaches to racial/ethnic differences in social determinants of mild cognitive impairment and its progression to dementia in the All of Us Research Program. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, *80*(12), gbaf179. https://doi.org/10.1093/geronb/gbaf179.

5. Sui, J.; Wu, B.; Zheng, Y.; Mo, Z.; **Dong, Q.**; Đoàn, L. N.; Yi, S. S.; and Qi, X. (2025). Racial and ethnic disparities in the burden of non-obese type 2 diabetes using different anthropometric measurements. *Obesity Medicine*, *53*, 100573. https://doi.org/10.1016/j.obmed.2024.100573.

6. **Dong, Q.**; Kline, D.; and Hepler, S. A. (2023). A Bayesian spatio-temporal model to optimize allocation of buprenorphine in North Carolina. *Statistics and Public Policy*, *10*(1), 2218448. https://doi.org/10.1080/2330443X.2023.2218448.

7. McKnight, E.; **Dong, Q.**; Brook, D. L.; Kline, D.; Hepler, S. A.; and Bonny, A. (2023). A descriptive study of opioid misuse prevalence and office-based buprenorphine access in Ohio: A call for a targeted and flexible approach. *Cureus*, *15*(3), e36903. https://doi.org/10.7759/cureus.36903.

Software

1. **surveyPrev** (R package): Mapping the Prevalence of Binary Indicators using Survey Data in Small Areas. (Author and maintainer)

2. **sae4health** (R package): An R Shiny Application for Small Area Estimation in Low- and Middle-Income Countries. (Author)

3. **sae4lmic** (website): A web-based application providing pre-calculated demographic and health estimates for selected countries and indicators. (Author)

Presentations

- Cross-fitting model evaluation for small area estimation using complex survey data. Poster and speed talk at JSM, Portland, OR, 08/24; Contributed talk at ENAR, Indianapolis, IN, 03/26; Topic-Contributed Paper Session at JSM, Boston, MA, 08/26

- A Bayesian spatio-temporal model to optimize allocation of buprenorphine in North Carolina. Poster presented at GEOMED, Irvine, CA, 10/22; Poster presented at OBAYES, Santa Cruz, CA, 09/22

- A latent-ordering partial order continual reassessment method (POCRM) with a distance-based prior for dose combination. Poster presented at SIP, Storrs, CT, 08/26; Poster scheduled at BBSW, Foster City, CA, 11/26

Teaching Experience

University of California, Santa Cruz **September 2022–Present**

Teaching Assistant:

- Courses: STAT 5: Statistics (Fall 2022), STAT 17: Statistical Methods for Business and Economics (Fall 2023, Spring 2024, Spring 2026), STAT 131: Introduction to Probability Theory (Winter 2024)

Wake Forest University **August 2020–May 2022**

Mathematics and Statistics Center Tutor:

- Courses: STA 111: Elementary Probability and Statistics, STA 368/668: Time Series and Forecasting.

Teaching Assistant:

- Courses: STA 111: Elementary Probability and Statistics, STA 212: Statistical Models, STA 362/662: Multivariate Statistics, STA 310: Probability.

Scholarships and Awards

- Dissertation-Year Fellowship, September 2026–June 2027
- Statistics Department Summer Fellowship, June 2023 and June 2026
- ICHPS 2023 Student Paper Travel Award, January 2023
- Regent Fellowship and Dean's Fellowship, August 2022–June 2023
- Full Tuition Scholarship, August 2020–May 2022
- Benjamin F. Marshall Memorial Scholarship, August 2018–May 2019
- R. Parker and M. S. Eastwood Scholarship, August 2017–May 2018
- Global Delegate Tuition Scholarship, August 2016–May 2019